

EFFECTIVENESS OF INPUT SUBSIDY PROGRAMS AMONG RURAL FARMERS IN MANGATAREM,PANGASINAN

An Undergraduate Thesis Presented to the Faculty
Members of the Economics Department
Pangasinan State University
Lingayen Campus
Lingayen, Pangasinan

In Partial Fulfillment of the
Requirements for the Degree
Bachelor of Arts in Economics

Caramat Irish I.
Pantaleon Rizza A.
Supangan Merry Angel M.
Vintacur Dennis F.
Evangelista Joseph M.

MARCH 2025

CHAPTER 1

INTRODUCTION

The preliminary portion of the study is presented in this chapter. Specifically, it shows the background of the study, the statement of the problem, the significance of the study, research hypothesis, and the scope and delimitation of the study.

Background of the Study

Agriculture is an important industry in the world economy, supplying food and income for millions of people. Agricultural input subsidies have long been used to promote smallholder farmers use of inputs, increase wages, reduce food prices, and promote economic growth (Crawford et al., 2003). Input subsidies mean giving farmers things like seeds, fertilizers, and pesticides at prices subsidized by the state. Public-sector farm investment involves investing in public goods such as irrigation systems, research and development facilities, and rural extension services (Zafar et al. 2023).

The most famous input subsidy scheme is the Malawi Fertilizer Subsidy Program, which took off in the early years of this century (Dorward et al., 2008). By offering smallholder farmers fertilizers at prices virtually below production costs, the project aimed to increase maize yields and thus the food security of millions of poor peasants in southern Africa. They indeed jumped substantially, but as with public investment programs like irrigation or power grids, it was rich farmers who benefited most. A number of critics argue that such a program depends for its existence upon external aid, suffers from malnutrition among beneficiaries, and is poor at penetrating through to the neediest peasant households, while many aspects of infrastructure

remain simply neglected and deteriorate. Its long-term prospects are nevertheless unclear.

Fertilizer subsidies were introduced by the government in Aceh Province of Indonesia to lower production costs and make it easier for farmers to obtain affordable agricultural inputs. These subsidies come from the state budget; they are distributed through various government agencies, state-owned enterprises, and private organizations to both improve and catalyze agricultural productivity (Yovo & Ganiyou, 2023). In Zambia, input subsidies represent the major expenditure in poverty reduction programs. However, maize yields are still below international benchmarks largely because the acidity of the soil curtails fertilizer efficiency. Researchers put forward that subsidies be expanded to cover soil additives such as lime in order to increase the effectiveness of fertilizers and productivity as a whole (Burke et al.).

To help farmers on one hand and solve the problem of food security on the other, China has introduced an agricultural subsidy policy since 2004. Seed subsidies, direct support to grain growers, and help paying for agricultural inputs finance constitute part of the program. This measure is taken to boost grain farming, cultivate food supplies, and raise farmers incomes (Li et al., 2022).

Due to sanctions imposed on Russia and the subsequent retaliatory measures undertaken by other countries, global fertilizer prices have rocketed up by a whopping 80% in 2022 (Paradis 2022). This put heavy pressure on farmers, especially in developing countries. In India, where agriculture employs half the national workforce and input prices are on the rise, already low farm incomes fell even further (Sodhi &

Tang 2014). Meanwhile, in China, more than 40 million peasants scrape by on less than (Zhao 2018).

The agricultural industry, an essential part of the world economy, includes the raising of crops and animals to support human existence and promote economic development (Dela Cruz et al. 1994; Aguilar, 2004; Antonio et al. 2009). Although it is important, agriculture encounters various challenges that include environmental problems, labor conflicts, and food safety issues (Esteban, 2008). Transparency stands out as an essential element in tackling these difficulties, promoting accountability, and building trust among stakeholders (Cruz, 2005). An innovative program designed to improve agricultural output and sustainability is the Fertilizer Voucher Scheme launched by the Department of Agriculture In the Philippines (Kusrini and Saleh, 2009). Initiated in 2011, this program aims to give small-scale farmers access to low-cost fertilizers via subsidized vouchers (Aquino et al., 2009). Although the program has demonstrated encouraging outcomes in boosting crop production and enhancing the well-being of farmers (Briones, 2020), issues like corruption and logistical obstacles have limited its impact (M. Onofrio, 2018).

The Department of Agriculture (DA), in collaboration with the Rice Resiliency Project 2 (RRP2), gave free fertilizer and high-quality rice seeds to farmers for the 2020-2021 dry season. The project comprised both RCEF and non-RCEF areas, with qualifying farmers receiving fertilizer vouchers of PHP2,000 to PHP3,000 per hectare. Despite these complaints, the DA had a distribution rate of more than 97% in Ilocos Norte, Ilocos Sur, and Isabela. The project aimed to help rice farmers during the pandemic and improve agricultural efficiency. (Department of Agriculture, 2021).The

Pangasinan government, through its Abig Pangasinan program, distributed PHP14.4 million in seeds and farm tools to farmers. The Department of Agriculture also provided PHP637.5 million in aid through the RCEF-RFFA program, which included cash awards, farm equipment, and animals. These projects demonstrate the governments efforts to increase agricultural output through input (subsidies. Farmers in Pangasinan Get P14-M Agri Inputs, n.d.).

The government would provide a one-time cash grant of PHP5,000 to 500 farmers in Mangatarem, Pangasinan, as part of the Rice Farmers Financial Assistance Program (RFFAP). Rice growers will receive aid via Land Bank cash cards, with PHP435 million put aside for 87,000 rice farmers in Pangasinan. The RFFAP Is one of the Rice Tariffication Law programs that aims to help farmers improve their competitiveness during the transition phase by providing financial support, mechanization, certified seeds, loans, and training(Gov't Grants P5-K Cash Aid for Pangasinan Farmers, n.d.).

Executive Order No. 60 restored the Philippine Crop Insurance Corporation (PCIC) to the Department of Agriculture (DA), allowing it to strengthen its crop insurance programs. It insured farmers against losses caused by natural disasters, pests, and diseases in order to sustain them. The reform will include provisions for increased farmer representation on the PCIC board, making it more farm-centered. It also aims to secure food, modernize agriculture, and encourage more Filipinos to farming(Da-Afid, 2024).

Mangatarem, a municipality in the province of Pangasinan, was chosen as the locale for this study due to its significant contribution to the province's agricultural

production. With a vast land area of 31,750 hectares, Mangatarem is one of the major agricultural producers in Pangasinan. The municipality's agricultural sector plays a vital role in the local economy, making it an ideal location to investigate the effectiveness of input subsidy programs among farmers.

Statement of the Problem

The purpose of this study is to evaluate the effectiveness of input subsidy programs among rural farmers in Mangatarem Pangasinan. Specifically, this study aims to answer the following:

1. What is the profile of the farmer respondents in terms of:

- a. Age
- b. Sex
- c. Civil Status
- d. Highest Educational Attainment
- e. Years of Farming Experience
- f. Farm Size and Location
- g. Type of Crops Cultivated
- h. Access to Government Subsidy program
- i. Other Sources of Income

2. What is the extent of effectiveness of the input subsidy programs in terms of:

- a. Cash assistance:
- b. Fertilizer subsidy program:
- c. Seed subsidy program:
- d. Crop insurance subsidy program:

3. What is the extent of effectiveness of the input subsidy programs in terms of:

- a. Income:

- b. Harvest:
- c. Yield:
- d. Cost of Production:
- e. Efficiency:
- f. Machinaries:
- g. Supply of Agricultural Crops:

4. If there a significant difference between the profile of farmers and the effectiveness of input subsidy programs among rural farmers in Mangatarem Pangasinan?

Research Hypothesis

In social sciences research, alpha represents the significance level, which is the probability of rejecting the null hypothesis. Researchers will test the hypothesis at different levels and will choose the lowest p-value approach. Since the significance level will be set at 0.05, the researchers will be 95 percent confident that the alternative hypothesis is valid and accepted.

Ho: There is no significant difference between input subsidy program and among rural farmers in Mangatarem, Pangasinan.

Significance of the Study

This section delineated an analysis of those who benefit from subsidies, thus showing the extent to which the subsidy policy benefits people and its effectiveness. First, the recipients are:

Farmers. Subsidies mainly make it easier for farmers to get cheaper seeds, chemical fertilizers, and pesticides. By reducing production costs, these plans can increase their income, raise crop yields, and strengthen their overall economic status. This study will evaluate whether subsidy programs lead to long-term development for farmers and instead help to among rural farmers.

Environment. Subsidy programs can positively and negatively impact the environment. They might encourage sustainable agriculture through the subsidization of composts and greener pesticides. But overuse of chemical inputs may create soil and water pollution problems. In this study that examine how to strike a balance between enhanced production and sustainability on the environment.

Market. Agricultural input industry is inter-correlated with agriculture sector and agricultural production and supply chain transactions. Greater production could firm up or bring prices down, making food more affordable. On the downside, subsidies can lead to dependence on government assistance or allow subsidized farmers to outcompete non-subsidized farmers. The project will study the effect of subsidies on market trends and whether they support the agricultural sector on a path of sustainable growth.

Rice Consumer. This is especially key as most consumers are indirectly benefiting from subsidization through enhanced food security and lower prices, especially for rice consumers. Farmers are able to produce more rice at lower costs, this is the only way to make rice affordable and available to many Filipinos, especially the low-income households. This will inform whether Delayed Input Subsidy Programs beneficially impact food pricing and stability of supply in the country.

Department of Agriculture. This study can help the Department of Agriculture assess the effectiveness of its rural farmer support programs. The studies conclusions will offer guidance for enhancing policy in order to make these initiatives more long-term viable and advantageous for farmers, particularly in Mangatarem, Pangasinan.

Policymaker. The findings can guide them in improving subsidy programs to better support farmers and ensure resources are used effectively. It will help in making informed decisions to maximize the programs' impact on rural farmers.

Future Researchers. The study offers a reference point for researchers in the future. It itself can also be the object of further research on agricultural subsidies and rural poverty reduction. Finally, it is something that provides inspiration for more studies into how to help farmers better and make subsidization programs more effective.

Scope and Delimitation of the Study

This research is being undertaken to determine whether farmer input subsidy programs are truly beneficial. We'll be looking at three major issues: how they effect farm productivity, household income, and living standards in Mangatarem, Pangasinan. Surveys with questionnaires will be undertaken among farmers who have received government payments for seeds, fertilizers, and pesticides. Cluster sampling will be used in this study to ensure that no ineligible farmers are included.

This work is limited to Mangatarem and excludes all other regions. It only covers input subsidies for seeds, fertilizers, and pesticides; however, it excludes support outlays such as financial aid, training, and the construction or alteration of irrigation

facilities. It also excludes much bigger economic factors, such as climate change or variations in market pricing that would affect level actual income for agricultural workers, as well as general agricultural policies. Because we are working with self-reported data, recall bias may alter the results. In addition, interviewers may provide subjective responses.

Definition of Terms

To provide clearer understanding for readers in this study, the following terms are defined operationally. The conceptual definition refers to the general meaning, while the operational definition pertains to the specific meaning within the context of this study.

Input Subsidy Program. It refers these are programs (either government or private sector) aimed to provide free or discounted seeds, fertilizers, pesticides, and machinery. In this Study Agricultural input subsidies have long been used to promote smallholder farmers use of inputs, increase wages, reduce food prices, and promote economic growth (Crawford et al. 2003).

Rural Farmers. Non urban areas of the Philippines depending on agriculture as their first source of income. In this paper they use the term beneficiaries to refer to participants of input subsidy programs that are being evaluated in terms of their agricultural production and economic livelihoods.

Cash Assistance. It refers a financial aid provided by the government or institutions to individuals without requiring repayment. In this study, it refers to support given to rice farmers in Mangatarem, Pangasinan under the Rice Farmers Financial

Assistance Program (RFFAP). This aid is sourced from rice import tariffs and distributed through Land Bank cash cards(Gov't Grants P5-K Cash Aid for Pangasinan Farmers, n.d.).

Fertilizer Subsidy. It refers distributing fertilizer so as to improve soil fertility and crop yield is called fertilizer distribution. In this study, it refers to the Rice Resiliency Project 2 (RRP2) of the Department of Agriculture(DA), which handed out free fertilizers to targeted rice farmers during the dry season of 2020-2021. The program used an e-voucher system, with eligible farmers allowed two to three bags of fertilizer for each hectare(Department of Agriculture, 2021).

Seed Subsidy. Agricultural assistance is the support provided to farmers in order to help improve their productivity and maintain stable economies of scale. In this study, it comprises seeds, tools and financial aid distributed under the Abig Pangasinan program and other initiatives of the Department of Agriculture. More specifically, farmers received certified seeds for rice, hybrid corn, farm tools and cash assistance to increase their harvest(Farmers in Pangasinan Get P14-M Agri Inputs, n.d.).

Crop Insurance. The Philippine Crop Insurance Corporation (PCIC) is a government-owned and controlled corporation (GOCC) that offers protection against loss of income to farmers caused by natural calamities, crop diseases or pest infestations in order to assure financial stability in agriculture. In this study, the PCIC refers to the corporation that was returned to the supervision of the Department of Agriculture (DA) May 13, 2024, through Executive Order 60, by insurance programs providing cover for small farmers against agricultural risks while ensuring effective policy coordination through broadly representative boards(Da-Afid, 2024).

Income. It refers the total money gained from all farm activities such as crop sales, government grants, and revenue from diversified farm activities (Todaro & Smith, 2020). In this Study measures if the input subsidy programs are one of the reasons that farmers' earnings increase by reducing the cost of the production process and by improving productivity.

Harvest. It refers is defined by FAO (Food and Agriculture Organization of the United Nations)as the process of collecting mature crops from fields after growing season has ended. This Study aims to examine the effect of input subsidies on the quantity and quality of harvests. If the amount of product garnered improves, this shows that program has increased output and efficiency in rural agriculture.

Yield. It refers to the amount of agro-ecological produce harvested per unit production area over a specific period and is expressed in kilogram or ton per hectare (World Bank, 2021). This research quantifies yield per hectare to ascertain the efficiency of input subsidy programs. The higher the yield, the better it is since farmers are enjoying subsidized inputs like fertilizers, seeds, and pesticides.

Cost in Production. It refers are all expenses related to farming alone, e.g seeds, fertilizers, pesticides, labor, every expense (Ellis, 2020). The research explores the extent to which input subsidies lower the overall cost of farming in Mangatarem, including seeds, fertilizer, pesticide, and labor expenses. A lower cost of production indicates the effectiveness of the program.

Efficiency. It refers indicates a rational allocation of resources, aiming to maximize output or yield and profit, and minimize cost and waste (Coelli et al.

2005).The study evaluates whether the subsidized inputs make farming more efficient by obtaining maximum output with minimal wastage of resources and cost of production.

Machienaries. Agricultural machinery is equipment that is mechanized and is used to perform agricultural tasks like tractors, plows, and harvesters to improve productivity (Singh,2019).This research examines whether input subsidy programs assist farmers in purchasing or accessing farm machinery, including tractors and harvesters, to enhance farm operation and productivity.

Supply of Agricultural Crops. The other is a sum of everything that farmers bring into the market. That sum will vary depending on farm output levels as well as the condition of wild, farmed fields both above ground and below it together with what governments do during this process which has been calculated in (FAO, 2021). Input subsidy programs create incentives for producers and traders in Mangatarem to invest more money and other resources into commodity production, and thus more products are grown. In a word, this will help ensure food security while creating a stable market situation.

Reference

- Bigting, K. A., Ramos, V., & Idava, P. (2024). The effectiveness of the fertilizer voucher scheme in general Tinio, Nueva Ecija. *The QUEST Journal of Multidisciplinary Research and Development*, 3(1). Retrieve from <https://doi.org/10.60008/thequest.v3i1.177>
- Dorward, A., & Chirwa, E. (2011). The Malawi Agricultural Input Subsidy Programme: 2005-6 to 2008-9. SSRN Eletronic Journal. Retrieve from [https://eprints.soas.ac.uk/9598/1/Malawi_fertilizer_case_-_Dorward %26 Chirwa Final R.pdfhttps://www.researchgate.net/publication/228285801_The_Malawi_Agricultural_Input_Subsidy_Programme_2005-6 to 2008-9](https://eprints.soas.ac.uk/9598/1/Malawi_fertilizer_case_-_Dorward_%26_Chirwa_Final_R.pdfhttps://www.researchgate.net/publication/228285801_The_Malawi_Agricultural_Input_Subsidy_Programme_2005-6_to_2008-9)
- Zafar, S., Aarif, M., & Tarique, M. (2023). Input subsidies, public investments and agricultural productivity in India. *Future Business Journal*, 9(1). Retrieve from <https://doi.org/10.1186/s43093-023-00232-1>
- Dorward, A., Roberts, P. D., Finegold, C., Hemming, D. J., Chirwa, E., Wright, H. J., Hill, R. K., Osborn, J., Lamontagne-Godwin, J., Harman, L., & Parr, M. M. (2013). Agricultural Input Subsidies for improving Productivity, Farm Income, Consumer Welfare and Wider Growth in Low- and Middle-Income Countries: A Systematic Review. scals ous Uploads Cabalpurita Scientists%20surmut Farmer
- The Effect of Agricultural input subsidies on Productivity: A Meta-Analysis. (2023). In Detox Development: Repurposing Environmentally Harmful Subsidies [Report]. Retrieve from <http://documents1.worldbank.org/curated/en/099432204102318322/pdf/IDU1e0e.0d35815c311449d1a9331ba19893a3664.pdf>
- Li, C., Sha, Z., Sun, X., & Jiao, Y. (2022b). The Effectiveness Assessment of Agricultural Subsidy Policies on Food Security: Evidence from China's Poverty-Stricken Villages. *International Journal of Environmental Research and Public Health*, 19(21), 13797. Retrieve from <https://doi.org/10.3390/ijerph192113797>
- Primalasari, I., & Octalia, V. (2024). ECONOMIC EFFICIENCY TO USE OF INPUT (SUBSIDIZED FERTILIZER) IN RICE FARMING IN MUSI RAWAS REGENCY (Case Study of Sentra Beras Sub-district). *Journal of Agri Socio Economics and Business*, 6(02), 211–226. Retrieve from <https://doi.org/10.31186/jaseb.6.2.211-226>
- ACB. 2016. Farm Input Subsidy Programmes (FISPs): A Benefit for, or the Betrayal of, SADC's Small-Scale Farmers? Johannesburg: ACB. Retrieve from https://www.acbio.org.za/wp-content/uploads/2016/07/Input-Subsidies_Report-ACBio.pdf

Gov't grants P5-K cash aid for Pangasinan farmers. (n.d.). Philippine News Agency. Retrieve from

<https://www.pna.gov.ph/articles/1089384>

Department of Agriculture. (2021, August 15). Distribution of fertilizers above board, benefiting targeted rice farmers. Official Portal of the Department of Agriculture. Retrieve from

<https://www.da.gov.ph/distribution-of-fertilizers-above-board-benefiting-targeted-rice-farmers/>

Farmers, fisherfolk in Pangasinan get P14-M agri inputs. (n.d.). Philippine News Agency. Retrieve from

<https://www.pna.gov.ph/index.php/articles/1159284>

Da-Afid. (2024, May 19). PBBM reverts Philippine Crop Insurance Corp. to DA. Official Portal of the Department of Agriculture. Retrieved from

<https://www.da.gov.ph/pbbm-reverts-philippine-crop-insurance-corp-to-da/>